

Oyendrila Dobe

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Education

- 2018 – 2024 ▶ **Ph.D. in Computer Science**
Michigan State University *East Lansing, USA*
Thesis: [Verification of Probabilistic Hyperproperties on Markov models](#).
Advisor: [Prof. Borzoo Bonakdarpour](#).
- 2013 – 2017 ▶ **B.Tech. in Computer Science and Engineering**
Maulana Abul Kalam Azad University of Technology *Kolkata, India*
Thesis title: *Detection and Extraction of Tumor from Brain MRI scans*.
Advisor: [Amiya Halder](#).

Work Experience

- 2024-current ▶ **Applied Scientist**
Team: Automated Reasoning Group, Amazon Development Center U.S., Inc. Boston, USA
- **Project:** *Kiro Specification Validation*
 - **Relevant Collaborator:** *Nadia Labai* and *Remi Delmas*
 - Involved in overall optimization of pipeline that detects ambiguity and inconsistency, in LLM-generated software requirements, and surfaces them to users for clarification.
 - Major contributions include designing of LLM-based refinement phase using abductive reasoning on the requirements, introduction of formalization-first approach instead of vocabulary-first to schema generation, auto-answering of clarification questions to reduce burden on users.
 - Blog link: [Deep spec analysis](#)
 - **Projects:** *Equalizer* and *JVerify*
 - **Relevant Manager:** *Anjali Joshi* and *Olivier Bouissou*
 - Worked on projects to tackle different aspects of verification of Java programs including deductive verification and equivalence checking.
 - Major contributions involved improving user experience on error reporting, additions of library contracts, building benchmarking toolkit for tracking improvement of equalizer, implementation of bitwise representation of integer handling in Java-to-Boogie translation.
- Summer 2023 ▶ **Applied Scientist Intern**
Team: Dafny, Amazon Development Center U.S., Inc. Seattle, USA
- **Project Manager:** *Aaron Tomb*; **Mentor:** *Rustan Leino*
 - Extended an existing tool that checks for cross-language equivalence of programs.

Work Experience (continued)

- Summer 2022 ▶ **Applied Scientist Intern**
Team: Privacy Engineering, Amazon Development Center U.S., Inc. New York, USA
- **Project Manager:** *Temesghen Kahsai*; **Mentor:** *Byron Cook*
 - Built a prototype tool to detect data residency violations in customers' cloud setup using automated reasoning techniques.
- 2020-2024 ▶ **Research Assistant**
Trustworthy and Reliable Technologies Lab, Michigan State University East Lansing, USA
- Advisor: *Prof. Borzoo Bonakdarpour*
 - Proposed solutions to various problems associated with the verification of probabilistic hyperproperties.
 - Build prototype tools to experiment with the proposed solutions.
 - Designed case studies and experiments to record and explain the results of the above tools on the case studies.
- 2018-2022 ▶ **Teaching Assistant**
Courses Taught: Object-Oriented Programming, Data Structures, Discrete Math, Computer Aided Verification (Graduate level)
- Conducted recitations, and office hours to help students improve their understanding.
 - Created and graded assignments and exams.
 - Lead TA for Fall 2022 for a class of 194 students.
- Summer 2019 ▶ **Research Assistant**
Program Analysis Laboratory, Iowa State University Ames, USA
- **Project Manager:** *Dr. Wei Le*
 - Built case studies to prove the applicability of demand-driven algorithm to generate minimum size executable of C++/C codes, from given program lines of large projects.
 - Wrote Python scripts to automate testing across docker containers and VM.
- 2017-2018 ▶ **System Engineer**
Infosys Limited Mysuru, India
- Underwent rigorous training in object-oriented programming in Python, and SAP ABAP.
 - Solely handled the implementation of leave system for a new subsidiary of the company using ABAP, directly under team-lead.
- Summer 2017 ▶ **Research Assistant**
Istaka Consulting Remote
- Conducted research on profiles of leading leadership coaches to form discussion agendas for one-on-one meetings.
 - Summarized and organized meeting notes, and assisted in website and presentation creation.

Research Publications

- 1 O. Dobe et al., "Lightweight verification of hyperproperties," in *Automated Technology for Verification and Analysis*, É. André and J. Sun, Eds., Cham: Springer Nature Switzerland, 2023, pp. 3–25, ISBN: 978-3-031-45332-8.
DOI: https://doi.org/10.1007/978-3-031-45332-8_1.

- 2 L. Gerlach, **O. Dobe**, E. Ábrahám, E. Bartocci, and B. Bonakdarpour, “Introducing asynchronicity to probabilistic hyperproperties,” in *Quantitative Evaluation of Systems*, N. Jansen and M. Tribastone, Eds., Cham: Springer Nature Switzerland, 2023, pp. 47–64, ISBN: 978-3-031-43835-6.  DOI: https://doi.org/10.1007/978-3-031-43835-6_4.
- 3 **O. Dobe**, E. Ábrahám, E. Bartocci, and B. Bonakdarpour, “Model checking hyperproperties for markov decision processes,” *Information and Computation*, vol. 289, p. 104 978, 2022, Special Issue on 11th Int. Symp. on Games, Automata, Logics and Formal Verification, ISSN: 0890-5401.  DOI: <https://doi.org/10.1016/j.ic.2022.104978>.
- 4 **O. Dobe**, L. Wilke, E. Ábrahám, E. Bartocci, and B. Bonakdarpour, “Probabilistic hyperproperties with rewards,” in *NASA Formal Methods*, J. V. Deshmukh, K. Havelund, and I. Perez, Eds., Cham: Springer International Publishing, 2022, pp. 656–673, ISBN: 978-3-031-06773-0.  DOI: https://doi.org/10.1007/978-3-031-06773-0_35.
- 5 **O. Dobe**, E. Ábrahám, E. Bartocci, and B. Bonakdarpour, “Hyperprob: A model checker for probabilistic hyperproperties,” in *Formal Methods*, M. Huisman, C. Păsăreanu, and N. Zhan, Eds., Cham: Springer International Publishing, 2021, pp. 657–666, ISBN: 978-3-030-90870-6.  DOI: https://doi.org/10.1007/978-3-030-90870-6_35.
- 6 E. Abraham, E. Bartocci, B. Bonakdarpour, and **O. Dobe**, “Parameter synthesis for probabilistic hyperproperties,” in *LPAR23. LPAR-23: 23rd International Conference on Logic for Programming, Artificial Intelligence and Reasoning*, E. Albert and L. Kovacs, Eds., ser. EPIc Series in Computing, vol. 73, EasyChair, 2020, pp. 12–31.  DOI: [10.29007/37lf](https://doi.org/10.29007/37lf).
- 7 E. Ábrahám, E. Bartocci, B. Bonakdarpour, and **O. Dobe**, “Probabilistic hyperproperties with nondeterminism,” in *Automated Technology for Verification and Analysis*, D. V. Hung and O. Sokolsky, Eds., Cham: Springer International Publishing, 2020, pp. 518–534, ISBN: 978-3-030-59152-6.  DOI: https://doi.org/10.1007/978-3-030-59152-6_29.
- 8 **O. Dobe**, A. Sarkar, and A. Halder, “Rough k-means and morphological operation-based brain tumor extraction,” in *Integrated Intelligent Computing, Communication and Security*, A. Krishna, K. Srikantaiah, and C. Naveena, Eds. Singapore: Springer Singapore, 2019, pp. 661–667, ISBN: 978-981-10-8797-4.  DOI: [10.1007/978-981-10-8797-4_67](https://doi.org/10.1007/978-981-10-8797-4_67).
- 9 A. Halder and **O. Dobe**, “Rough k-means and support vector machine based brain tumor detection,” in *2017 International Conference on Advances in Computing, Communications and Informatics (ICACCI)*, 2017, pp. 116–120.  DOI: [10.1109/ICACCI.2017.8125826](https://doi.org/10.1109/ICACCI.2017.8125826).
- 10 A. Halder and **O. Dobe**, “Detection of tumor in brain mri using fuzzy feature selection and support vector machine,” in *2016 International Conference on Advances in Computing, Communications and Informatics (ICACCI)*, 2016, pp. 1919–1923.  DOI: [10.1109/ICACCI.2016.7732331](https://doi.org/10.1109/ICACCI.2016.7732331).

Miscellaneous Experience

Service - Peer review

Program Committee	► <i>EMSOFT 2025, SpecOps 2026</i>
SubReviewer	► <i>CAV 2024, EMSOFT 2024, VSTTE 2023, NFM 2022, QEST 2022, DISC 2022, ATVA 2021</i>
Artifact Evaluator	► <i>TACAS 2026, 2025, 2024, CAV 2025, 2024, OOPSLA 2026, iFM 2025, QEST 2023, 2022, PLDI 2023</i>